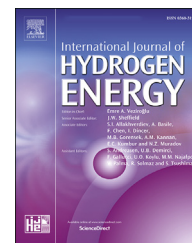


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Experimental study of porosity and effective thermal conductivity in packed bed of nano-structured FeTi for usage in hydrogen storage tanks

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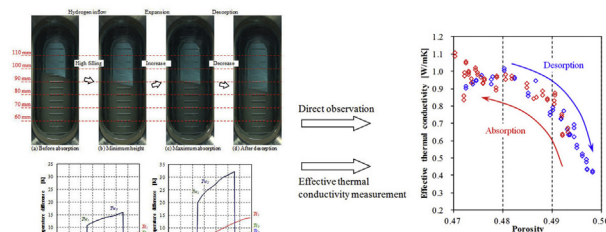
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HIGHLIGHTS

- Nano-structured FeTi (n-FeTi) has excellent performance as a hydrogen storage alloy.
- The volume of n-FeTi packed bed decreased by the cycles of hydrogenation.
- The effective thermal conductivity (ETC) in packed bed of n-FeTi was measured.
- ETC of n-FeTi increased with reacted fraction rise and porosity decrease.
- Manufacturing hydrogen storage tanks packed n-FeTi can be put to practical use.

GRAPHICAL ABSTRACT



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ABSTRACT

The porosity and effective thermal conductivity, which were measured simultaneously, changed owing to expansion during hydrogen absorption and contraction during hydrogen desorption. The metal hydride used in the experiment was a newly developed nano-structured FeTi (n-FeTi) by mechanical alloying. The porosity was measured by directly observing the changes in the volume of a metal hydride packed bed under free expansion. In the authors' previous study of LaNi₅, the volume of the packed bed gradually increased, whereas the total volume of the n-FeTi packed bed decreased with increase in the cycles of expansion and contraction. The effective thermal conductivity measurements of the metal hydride packed bed were carried out using a cylindrical device with heating from a central

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heater. The results showed that the effective thermal conductivity of n-FeTi changes from 0.4 to 1.1 W/mK with rise in reacted fraction and decrease in porosity.

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Introduction

The problem with most renewable energy sources is unsteady supply. For steady source of electrical power, its storage is essential. Hydrogen is an ideal source of electricity because it can be stored for long-term and on large scales. Hydrogen can be obtained by subjecting fresh water to an electrolytic process. Moreover, when hydrogen generates electricity by a fuel cell, it can mitigate the emission of greenhouse gases because the only product of fuel cell is water [1,2].

The ability to store hydrogen is considered as the main problem in the usage of hydrogen as a source of energy [3]. Metal hydrides have many benefits for hydrogen storage [4–7]. Particularly, metal hydrides tanks are promising candidates for safely storing hydrogen because they do not easily explode. Metal hydride tanks can store hydrogen in high volumes at moderate temperatures and pressures. For practical hydrogen storage applications, the basic thermodynamic and kinetic properties of metal hydrides should be studied more closely. Experiments [8], analysis [9] and reviews [10,11] were conducted for practical tanks. Furthermore, stress is important factor to consider when designing and using the metal hydride tanks [12–15].

The hydrogen absorption/desorption of metal hydrides is an exothermic/endothermic reaction. Therefore, heat transfer enhancement of metal hydrides is essential for achieving a continuous process [16–18]. For design of practical hydrogen tanks, a numerical analysis of the thermal conductivity of metal hydride used in powder form is essential. In recent years, many proposals for mathematical models [19–22] of metal hydride reactors have been made, and Mohammadshahi et al. [23] presented a review. The effective thermal conductivity (ETC) is the thermal conductivity of the entire packed bed. For other simple powders, the effective thermal conductivity of a packed bed changes with the packing structure of powder [24,25]. For reactive packing beds of metal hydride powders, the packing structure is more important [26]. The packing structure has many parameters [27,28]. Porosity is also an important parameter [27]. The relationship between the porosity and the expansion and contraction of the entire packed bed is essential for tank development [29–31]. Direct observation is effective in examining changes in the packed bed [32,33]. Therefore, the authors measured the porosity of a metal hydride LaNi₅ packed bed by direct observation under free expansion [32]. The experimental results showed that the porosity of the packed bed gradually increased with increase in hydrogen absorption and desorption cycles. The porosity increase of LaNi₅ is not preferable for hydrogen storage tanks.

FeTi is an attractive candidate for practical use because it is relatively inexpensive and has a high storage capacity [34]. However, FeTi has a disadvantage, i.e., the initial activation is

difficult. Recently it was shown that, mechanical alloying by ball milling can be used to form nanostructured materials that are very different from traditional crystalline materials [35]. It has been reported that nano level crystals are an improvement in hydrogen storage characteristics [36]. Thus, nano-structured FeTi (n-FeTi) formed by mechanical alloying is attractive. The n-FeTi has fine grain size, exhibits high initial rate of hydrogen absorption and limited particle breakup and degradation. The authors reported only the direct observation of volume changes of n-FeTi packed bed in the previous study [37].

In this work, the porosity and the effective thermal conductivity, that were measured simultaneously, changed owing to the influence of the expansion during hydrogen absorption and contraction during hydrogen desorption. It is important to measure both the parameters at the same time. The porosity was measured by directly observing the volume changes of a metal hydride packed bed under free expansion, whereas the effective thermal conductivity measurements of a metal hydride packed bed were carried out by a cylindrical device with heating from a central heater. An analytical method using Laplace transformation developed by Monde et al. [38] was used for experimental data processing. Using this method, it is possible to measure the effective thermal conductivity of the metal hydride during hydrogen absorption.

The purpose of this paper is to show the effect of hydrogen absorption on effective thermal conductivity and the effect of porosity obtained by direct observation experiment on effective thermal conductivity. Furthermore, this articles shows that hydrogen storage tank manufactured by n-FeTi is very practicable.

Experimental apparatus and methods

Experimental apparatus

The schematic diagram of the experimental apparatus is shown in Fig. 1. The experimental apparatus consists of a hydrogen supply system, a pressure chamber, and a circulating water system. Hydrogen was stored in the secondary tank (304L-HDF4-1000; –500; Swagelok, USA) to calculate the amount of hydrogen storage. High purity (99.999% or more) hydrogen is used, and the impurities in hydrogen have little effect on the pulverization behavior. The pressure chamber in the test section was kept at uniform temperature by circulating water (MCX350, AS ONE Corporation, Japan). Fig. 2 shows the photo and detailed diagram of test section. The observation pressure chamber is a reaction vessel with glass windows and is made of sight glass (S8X, Daido Machines Co., Ltd., Japan) and stainless-steel flanges. Maximum pressure that the observation pressure chamber can withstand is about 1.0 MPa.

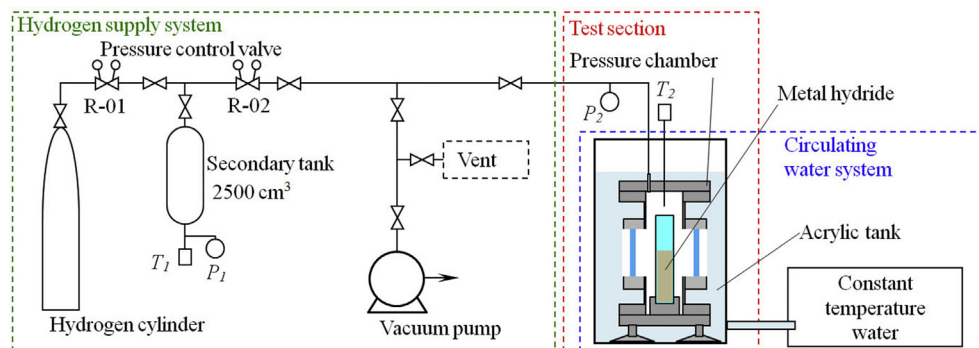


Fig. 1 – Schematic diagram of the experimental apparatus.

The detailed diagram of the internal glass container of metal hydride is shown in Fig. 3. The internal glass diameter of the internal glass container is 26 mm. The internal glass container has a heater in the center for effective thermal conductivity measurements. The heat flux can be arbitrarily changed by controlling the direct current to the heater (H35K, Okazaki Manufacturing Company, Japan). Therefore, by constant heating at a constant voltage, it is possible to conduct a heat conduction experiment in a cylindrical coordinate system of constant heat flux. Length of the heating part is 50 mm, and the heater cover diameter is 6.35 mm. The temperature changes were measured with four thermocouples placed at different distances from the center (mid height) of the heating part. Fig. 3(b) shows the cross section including the positions of the thermocouples.

The pressures of the secondary tank (P_1) and of the pressure chamber (P_2) were measured using absolute manometers (P_1 : NS100A-5MPa; P_2 : NS100A-2MPa, Minebea Co., Ltd., Japan). The temperatures of the secondary tank (T_1), the test chamber (T_2), the metal hydride packed bed (T_{c1} , T_{c2} , T_{c3} , T_{c4}),

the acrylic tank water, and the circulating water were measured using K-type thermocouples (AD-THERMIC, Yamari Industries Limited, Japan). The pressure and temperature data were measured every 0.5 s and were recorded by a measurement hardware (8423, HIOKI E.E. Corporation, Japan).

Experimental methods

Activated metal hydride was transferred to the internal glass container of the pressure chamber in a helium atmosphere. The weight of the metal hydride put into the internal glass container was 128 g. The uniform temperature of the circulating water was 25.0 °C. It has been reported that n-FeTi prepared by mechanical alloying shows the first plateau pressure at about 0.5 MPa in hydrogen gas at 25.0 °C [36]. In the present study, the pressure was limited to below 1.0 MPa, because practical hydrogen tanks are designed at below 1.0 MPa. Thus, the experiments were carried out in first plateau pressure region and did not include the second plateau pressure region.

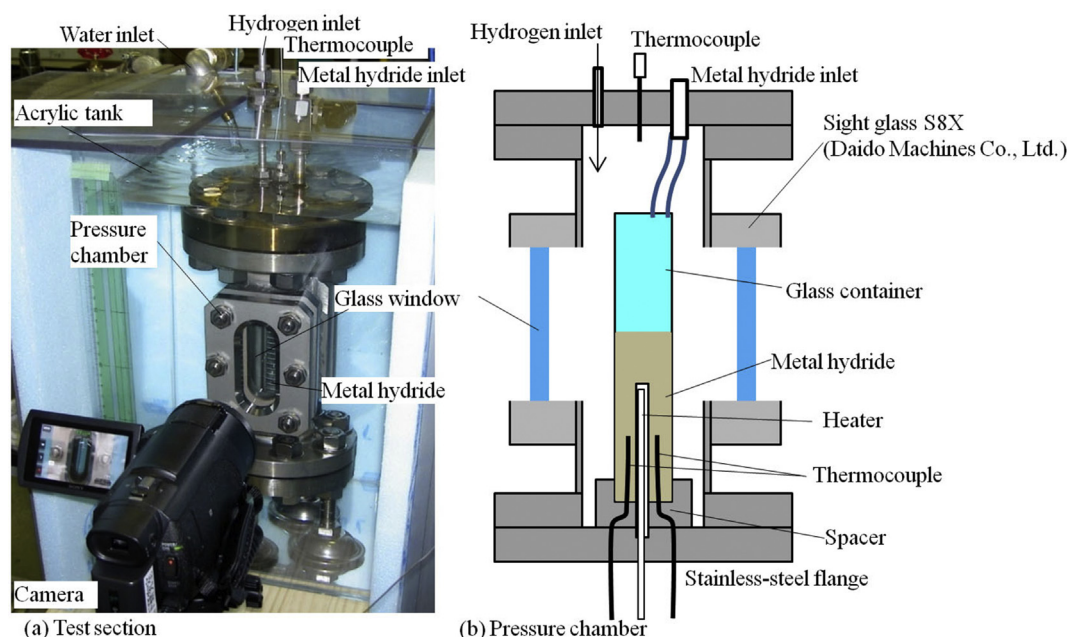


Fig. 2 – Photo and detailed diagrams of test section.

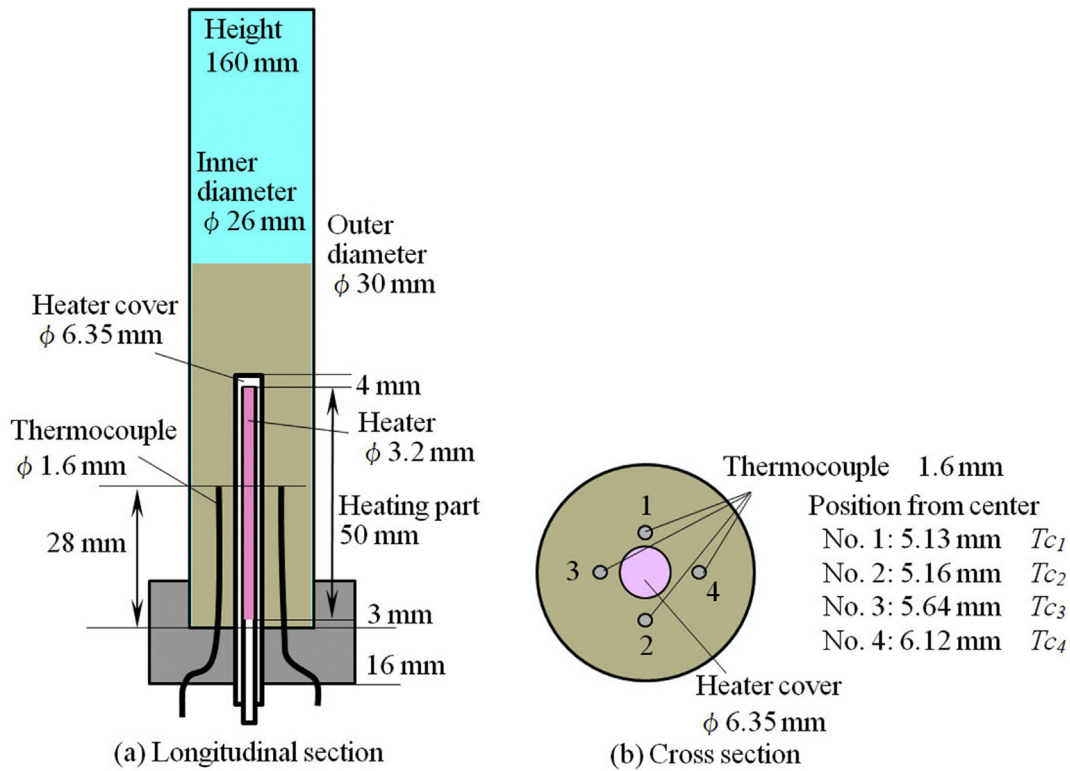


Fig. 3 – Detailed diagrams of glass container.

For hydrogen absorption, hydrogen was gradually supplied until the pressure of P_2 exceeded 0.98 MPa. The completion of absorption was judged by the difference between metal hydride temperature and circulating water temperature. Then, the metal hydride was sustained in constant pressure and equilibrium temperature for several hours to fully absorb.

For hydrogen desorption, hydrogen was slowly removed from the packed bed by the vacuum pump (GLD-051, ULVAC-RIKO, Inc., Japan). The pressure chamber was evacuated over several hours, and then the metal hydride was held in the vacuum over several hours until the pressure was below 10 Pa.

Absorption and desorption cycle was repeated 26 times. The metal hydride-stored hydrogen was about 1.0 mass% at the end of absorption. In this paper, the reacted fraction F , defined as the ratio of reacted hydrogen to the maximum hydrogen storage amount at 0.98 MPa is used instead of the weight ratio.

Direct observation

Experimental data processing for direct observation are similar to previous studies [32,37]. The volume change of the metal hydride packed bed was recorded using a video camera (FDR-AX55, Sony Corporation, Japan). The internal glass container has gauge lines. The mean height of the packed bed was calculated using the top boundary distribution.

The expansion ratio was simply calculated from each volume:

$$\phi_{ab} = \frac{(V_1 - V_0)}{V_0} \quad (1)$$

where V_0 is the minimum volume of a cycle, and V_1 is the maximum volume of a cycle.

Porosity was calculated by the following formula:

$$\varepsilon = 1 - \frac{M(F)}{V \times \rho(F)} \quad (2)$$

where M is the weight of packed metal hydride, V is the volume of the packed bed, and ρ is the solid density of the metal hydride. The variables M and ρ are calculated to vary linearly with hydrogenation. This paper deals with the average porosity of the entire packed bed.

Effective thermal conductivity measurement

The analytical method using Laplace transform developed by Monde et al., [38] was used for experimental data processing.

For a finite material, to estimate the surface temperature using the inverse solution, two measured temperatures are required. The heater surface temperature T_{w1} was calculated by temperature of two different positions (T_{c1} , T_{c2}). The determination of the effective thermal diffusivity and the effective thermal conductivity requires another surface temperature T_{w2} calculated by temperature of two different positions (T_{c1} , T_{c3}). The effective thermal diffusivity and the effective thermal conductivity must be determined to satisfy the following equation.

$$E_r(\alpha, \lambda) = \int_{t_1}^{t_2} (T_{w1}(t, \alpha, \lambda) - T_{w2}(t, \alpha, \lambda))^2 dt \rightarrow 0 \quad (3)$$

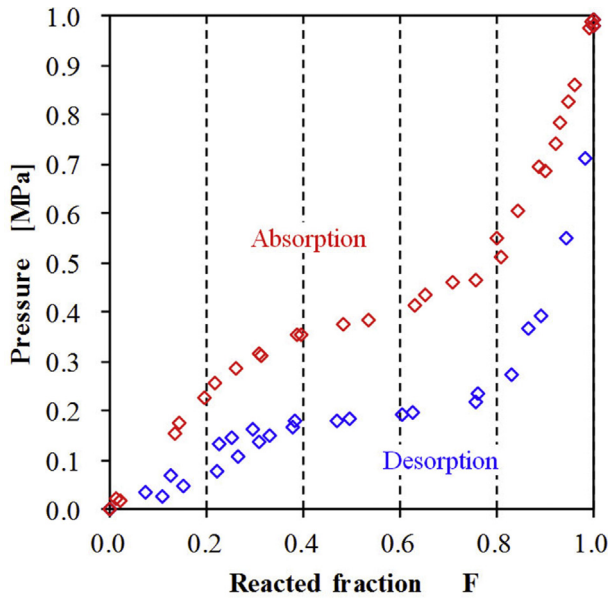


Fig. 4 – Relation between reacted fraction and pressure at effective thermal conductivity measurements.

where α is the effective thermal diffusivity and λ is the effective thermal conductivity.

Therefore, at least three measured temperatures are required. The temperature change was measured with four thermocouples with different distances from the center. The result is that both effective thermal diffusivity and thermal conductivity can be obtained, but this paper reports only effective thermal conductivity.

The effective thermal conductivity measurements were carried out from 25 to 26 cycles. The measurements were taken at perfect equilibrium after several hours at constant pressure. Fig. 4 shows the relationship between the reacted

fraction and the pressure in the effective thermal conductivity measurements. The applied voltage was constant at 3 V, and then the heat flux was about 6.6 kW/m^2 . The heating time was 90 s.

Experimental results and discussion

Direct observation results

Fig. 5 shows the volume change images of the n-FeTi packed bed at first absorption/desorption cycle. Fig. 5(a) shows the volume before absorption. Just after hydrogen flows in, the volume is minimum [Fig. 5(b)]. At maximum hydrogen absorption, the volume is maximum [Fig. 5(c)]. The hydrogen desorption reduces the volume but does not reduce it completely. At the end of a cycle, the packed bed is loose [Fig. 5(d)]. The decrease in volume owing to hydrogen inflow and the loose state of the packed bed after desorption occurred during every cycle.

Fig. 6 shows volume comparison at maximum absorption in each cycle. The result shows that volume gradually decreased with every passing cycle, and the volume change almost stopped at about 10 cycles. In case of LaNi_5 [32], volume increased with every passing cycle because of particles breakup. Owing to particle breakup, the particle diameter sizes decrease and become uniform, and the particles' surfaces become rough. Then, the packed bed could not be packed at high density, and the volume gradually increased. Whereas, in case of n-FeTi, the volume of packed bed gradually decreased, because particle breakup was less. Moreover, the volume changes stopped at a relatively earlier cycle.

Expansion ratio and porosity

The expansion ratios are calculated using Eq. (1). The changes in the expansion ratios with respects to cycle number is

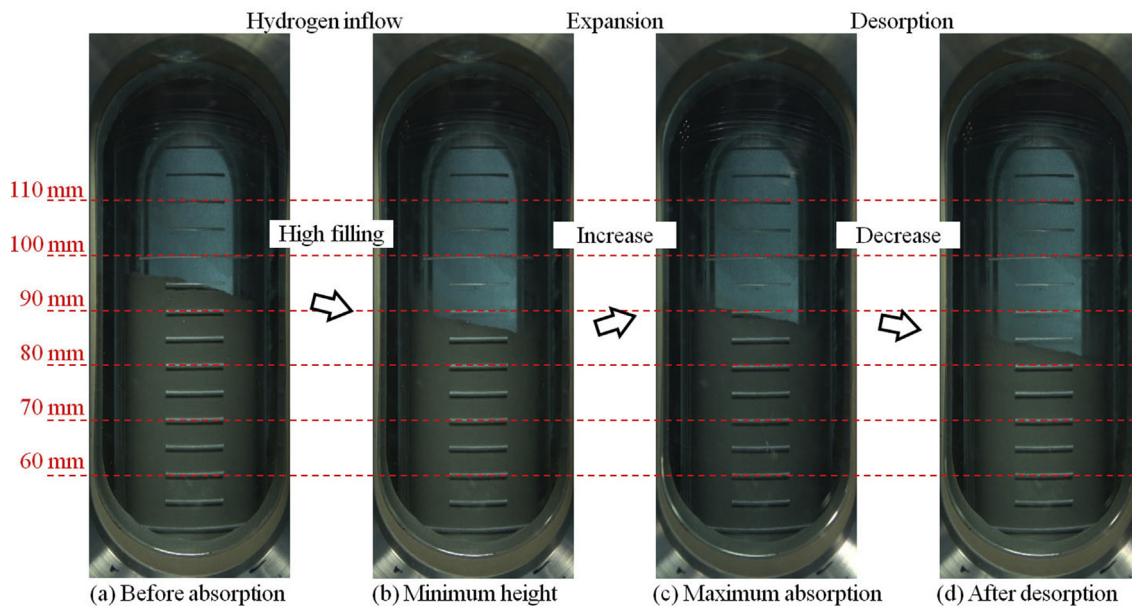


Fig. 5 – Change in volume of n-FeTi hydrogen storage alloy at first absorption-desorption cycle.

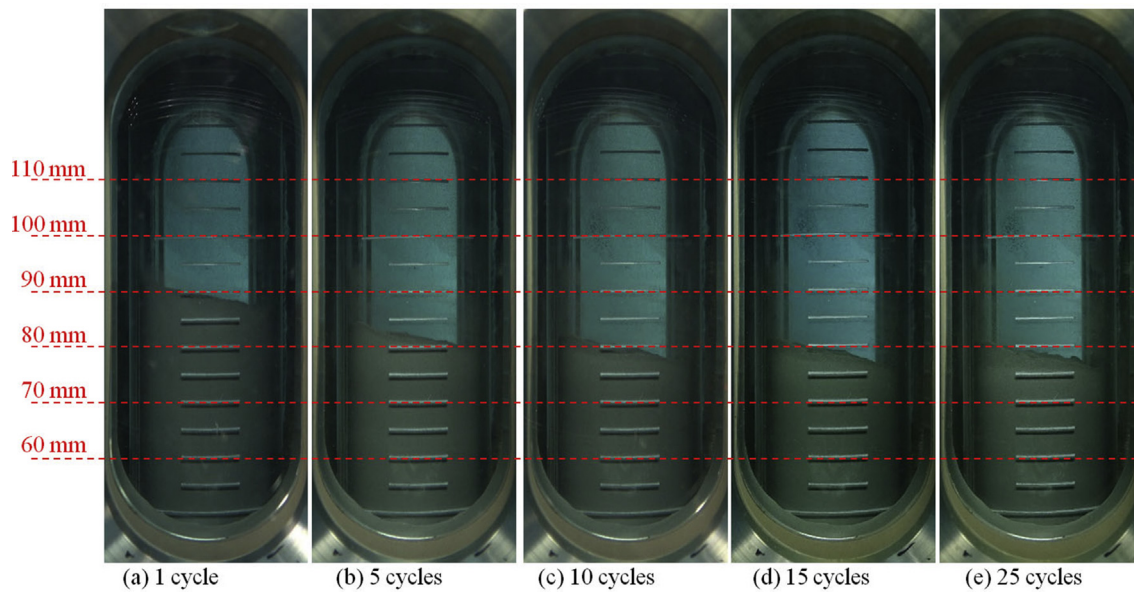


Fig. 6 – Change in volume of n-FeTi hydrogen storage alloy at maximum absorption by repetition of hydriding.

shown in Fig. 7. There is no significant influence by the heater. Although the particle expansion ratio of n-FeTi is 10.5%, the expansion ratio of n-FeTi overall packed bed is 5% or less. The results of the expansion ratio of LaNi₅ packed bed [32] are also included in Fig. 7. The expansion ratio is affected by L/D which is the ratio of the packed bed depth to the inner diameter. Therefore, Fig. 7 also shows the value of L/D. Although experiments on the detail influence by L/D shall be future task for the authors, the expansion ratio of n-FeTi packed bed owing to hydrogen absorption/desorption is evidently smaller than LaNi₅. A low expansion ratio of n-FeTi is preferred for hydrogen storage tank.

The porosity of the packed bed calculated by Eq. (2) is shown in Fig. 8. The porosity using the maximum volume V_1 is ϵ_1 , and ϵ_2 is the porosity using the volume V_2 after desorption at the end of a cycle. All porosity of n-FeTi in Fig. 8(a) decreased contrary to LaNi₅ in Fig. 8(b). The change in porosity stopped after 10 cycles for n-FeTi. Whereas, the change in porosity continued at about 30 cycles in LaNi₅. The porosity of LaNi₅ exceeded 50%, but the porosity of n-FeTi was 50% or less. This difference in porosity change could be due to the difference in the particle breakup of the two metals.

The limit porosity of the packing is mainly influenced by two factors: particle shape and particle size distribution. In

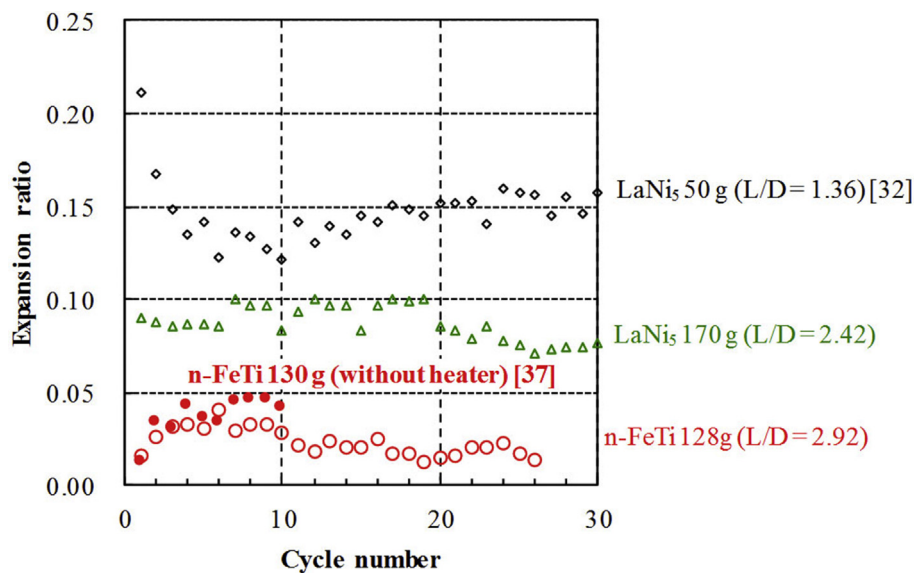


Fig. 7 – Expansion ratio as a function of cycle number. Red hollow circle symbols are the results of this experiment. Red filled circle symbols are the results of observation experiment without heater [37]. Black hollow diamonds indicate the expansion ratio of LaNi₅ packed bed [32].

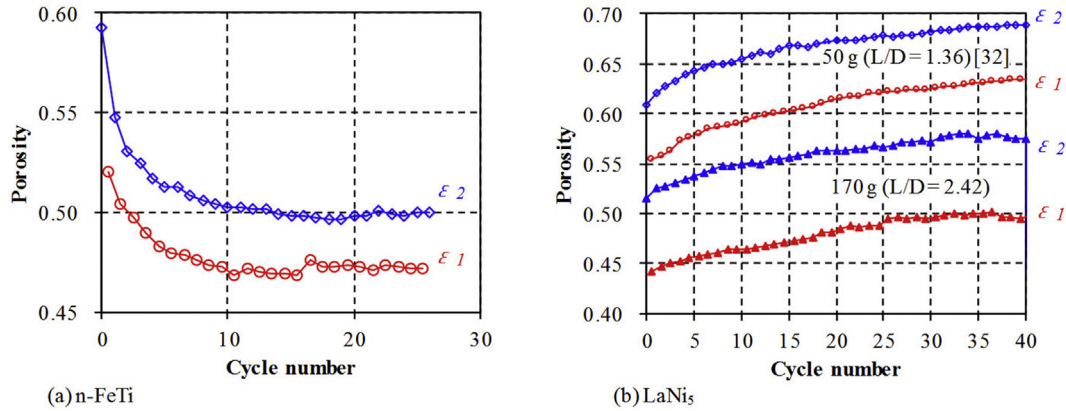


Fig. 8 – Porosity as a function of cycle number.

case of LaNi₅ of the previous paper [32], the mean particle diameter at the initial stage is about 103.6 μm , and particle size distribution has no peak. The particles rapidly broke up until 15 cycles of absorption and desorption. Then, the mean particle diameter is about 17.4 μm , and the peak of the particle size distribution is about 12.9 μm . Due to the particle breakup, the particle surface becomes rough and the particle size distribution becomes uniform. Therefore, the particles breakup increases the limit porosity of packing. After 15 cycles, the whole volume of the packed bed gradually increased. In case of n-FeTi formed by MA in this experiment, the mean particle diameter at the initial stage is about 29 μm of fine grain size. The particles exhibits little particle breakup by cycles of absorption and desorption. The porosity of the packed bed of n-FeTi decreased by every passing cycle due to decrease the initial packing looseness. The change in porosity stopped at early cycle about 10 cycles. And after 10 cycle, the porosity of n-FeTi was continuously stable. The stability porosity of n-FeTi is preferred for hydrogen storage tanks.

Effective thermal conductivity

Fig. 9 shows the examples of the temperature difference histories in the maximum hydriding case and in the low hydriding case. In many measurements, because T_{c2} and T_{c3} were nearly the same, only the temperature changes of T_{c1} , T_{c2} and T_{c4} are shown in the figure. The heater surface temperature T_w was calculated using temperature of two different positions. T_{w1} was calculated from T_{c1} and T_{c2} , T_{w2} was calculated from T_{c1} and T_{c4} . Since using the entire temperature measurement of 90 s does not produce accurate results, data processing were performed a part time intervals. The start of the data processing was set to be when a change of 1.0 K has occurred at all temperature measurement points. The process interval was set to about 50 s (about 100 data). The effective thermal conductivity for which the two surface temperatures were same, was searched. The surface temperatures that became nearly the same are shown in Fig. 9. During data processing, heat flux was also calculated

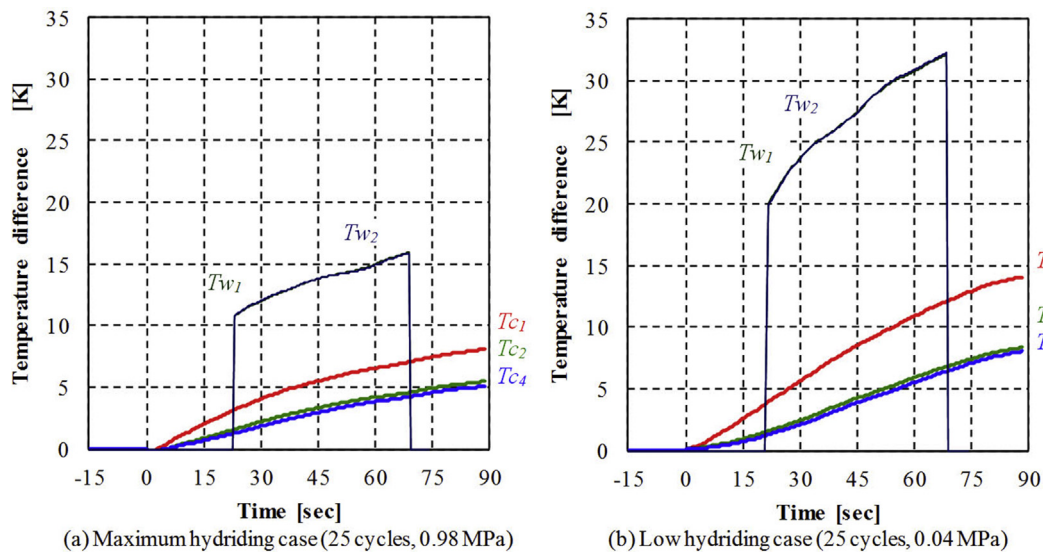


Fig. 9 – Examples of temperature difference histories.

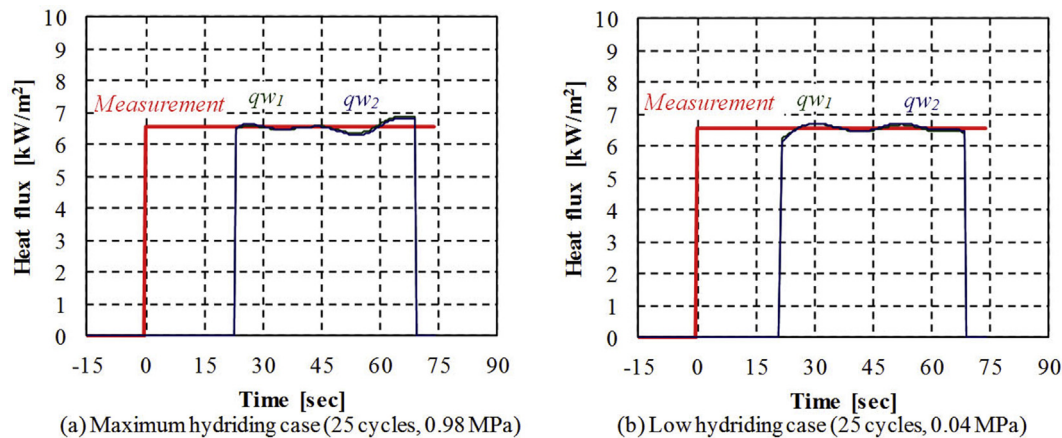


Fig. 10 – Examples of heat flux histories.

simultaneously. Fig. 10 shows the examples of the heat flux histories. The heat flux q_{w1} was calculated together when Tw_1 is calculated from T_{c1} and T_{c2} . Similarly, q_{w2} was calculated together with Tw_2 . The two heat fluxes overlap and the measured values almost match. The obtained effective thermal conductivities were 1.11 W/mK for maximum hydriding case and 0.43 W/mK for low hydriding case, respectively.

Fig. 11 shows the relation between the reacted fraction and the effective thermal conductivity from 25 to 26 cycles. The effective thermal conductivity of n-FeTi changes from 0.4 to 1.1 W/mK according to reacted fraction rise. The results are compared with experiments of $LaNi_5$ that were reported by Yoshida et al. [39]. The thermal conductivity of metal hydride is unknown, and should be different for $LaNi_5$ and n-FeTi. Moreover, the porosity of $LaNi_5$ and n-FeTi was different. Therefore, the effective thermal conductivity is considered to be different, however, both thermal conductivities show a similar range and

the same trend fortuitously. The effective thermal conductivity of n-FeTi increases with increase in reacted fraction.

Fig. 12 shows the relation between the porosity and the effective thermal conductivity from 25 to 26 cycles. The effective thermal conductivity of n-FeTi increases with decrease in porosity. The effective thermal conductivity is predictable if the reacted fraction or porosity are determined. The influence of the porosity is clearly shown in Fig. 12. However, for the pressure, since the reacted fraction and the porosity are different between at absorption and at desorption, the effective thermal conductivity cannot be determined. The relation of porosity, reacted fraction and effective thermal conductivity of the metal hydride packed bed obtained in this experiment is considered to be useful for the numerical analysis of n-FeTi and design of practical hydrogen tanks.

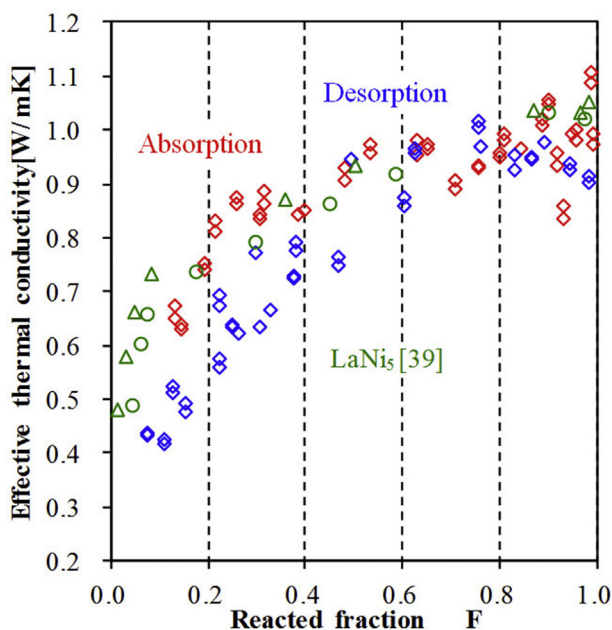


Fig. 11 – Relation between reacted fraction and effective thermal conductivity.

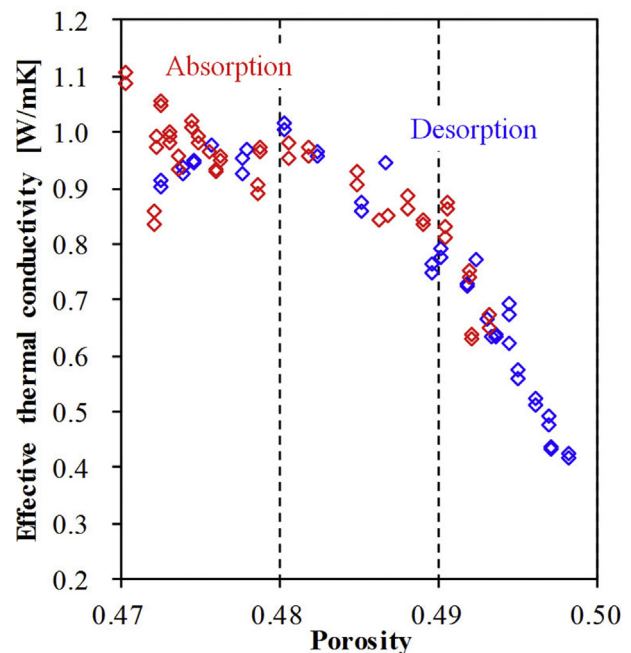


Fig. 12 – Relation between porosity and effective thermal conductivity.

Conclusions

This paper showed the experimental results of direct observation of volume changes and the effective thermal conductivity measurements of packed bed nano-structured FeTi (n-FeTi) that was developed by mechanical alloying. The porosity and effective thermal conductivity changed owing to the influence of expansion during hydrogen absorption and contraction during hydrogen desorption, which were measured simultaneously.

The porosity was measured by directly observing the changes in volume of a metal hydride packed bed under free expansion. The experimental results showed that the total volume of the n-FeTi packed bed decreased with every passing cycle of expansion and contraction unlike LaNi₅. This difference in the porosity change could be due to the difference in particle breakup of the two metals. The reduced porosity of n-FeTi is preferred for hydrogen storage tanks.

The effective thermal conductivity measurements of a metal hydride packed bed were carried out by a cylindrical device with heating from a central heater. An analytical method using Laplace transformation was used for experimental data processing. The results showed that the effective thermal conductivity of n-FeTi changes from 0.4 to 1.1 W/mK according to reacted fraction rise and according to porosity decrease.

The relation between porosity, the reacted fraction and effective thermal conductivity of the metal hydride packed bed obtained in this experiment is considered to be useful for the numerical analysis of n-FeTi and design of practical hydrogen tanks.

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